



Instituto Tecnológico Nacional de México

Instituto Tecnológico de Saltillo



*Cómputo en la Nube*

Miguel Salazar del Bosque

#### **4.3 Seguridad(hardening)en servidor web Apache 2**

Roberto Aram López Rodríguez

No. Control: 22050788

20-Marzo- 2026

## Introducción

En esta práctica se realizó la configuración de un servidor Apache aplicando técnicas de hardening con el objetivo de mejorar la seguridad del sistema, reduciendo riesgos y protegiendo la información del servidor.

## Desarrollo

### Paso 1. Apache funcionando

```
roberto@Debian-Server:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
   Active: active (running) since Fri 2026-03-20 07:27:21 CST; 6min ago
     Invocation: efeb1c0dfa0f405b854e15fd2733d2dc
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 3609 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SI
   Main PID: 3612 (apache2)
      Tasks: 7 (limit: 2297)
     Memory: 13.8M (peak: 14M)
        CPU: 158ms
    CGroup: /system.slice/apache2.service
            └─3612 /usr/sbin/apache2 -k start
              └─3614 /usr/sbin/apache2 -k start
                └─3615 /usr/sbin/apache2 -k start
                  └─3616 /usr/sbin/apache2 -k start
                    └─3617 /usr/sbin/apache2 -k start
                      └─3618 /usr/sbin/apache2 -k start
                        └─3622 /usr/sbin/apache2 -k start
```

```
Mar 20 07:27:21 Debian-Server systemd[1]: Starting apache2.service - The Apache
Mar 20 07:27:21 Debian-Server systemd[1]: Started apache2.service - The Apache
lines 1-21/21 (END)
```

Se verificó que el servidor Apache se encuentra activo y funcionando correctamente.

### Paso 2. Página en navegador



## It works!

This is the default web page for Apache.

Se comprobó el funcionamiento del servidor accediendo desde el navegador, mostrando la página predeterminada de Apache.

### Paso 3. Hardening (ocultar info)

---

```

# Changing the following options will not really affect the security of the
# server, but might make attacks slightly more difficult in some cases.

#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
[ Read 58 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut      ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify   ^_ Go To Line
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature Off

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On

#
# Forbid access to version control directories

```

```

^G Help      ^O Write Out ^F Where Is  ^K Cut      ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify   ^_ Go To Line

```

Se modificó el archivo de configuración para ocultar información sensible del servidor y mejorar la seguridad.

## Paso 4. Desactivar módulos

```
roberto@Debian-Server:~$ sudo apache2ctl -M
```

Loaded Modules:

```
core_module (static)
so_module (static)
watchdog_module (static)
http_module (static)
log_config_module (static)
logio_module (static)
version_module (static)
unixd_module (static)
access_compat_module (shared)
alias_module (shared)
auth_basic_module (shared)
authn_core_module (shared)
authn_file_module (shared)
authz_core_module (shared)
authz_host_module (shared)
authz_user_module (shared)
autoindex_module (shared)
deflate_module (shared)
dir_module (shared)
env_module (shared)
filter_module (shared)
mime_module (shared)
mpm_prefork_module (shared)
negotiation_module (shared)
php_module (shared)
reqtimeout_module (shared)
setenvif_module (shared)
status_module (shared)
```

```
roberto@Debian-Server:~$ sudo a2dismod autoindex status proxy proxy_http
```

[sudo] password for roberto:

WARNING: The following essential module will be disabled.  
This might result in unexpected behavior and should NOT be done  
unless you know exactly what you are doing!  
autoindex

To continue type in the phrase 'Yes, do as I say!' or retry by passing '-f': Yes  
, do as I say!

Module autoindex disabled.

Module status disabled.

Module proxy already disabled

Module proxy\_http already disabled

To activate the new configuration, you need to run:

```
systemctl restart apache2
```

```
roberto@Debian-Server:~$ sudo systemctl restart apache2
```

Se deshabilitaron módulos innecesarios para reducir la superficie de ataque del servidor.



## Paso 5. HTTPS (LO MÁS IMPORTANTE)

```
roberto@Debian-Server:~$ sudo apt install openssl ssl-cert -y
sudo a2enmod ssl
ssl-cert is already the newest version (1.1.3).
ssl-cert set to manually installed.
Upgrading:
  libssl3t64  openssl  openssl-provider-legacy
```

Summary:

```
Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 92
Download size: 0 B / 4,252 kB
Space needed: 15.4 kB / 10.8 GB available
```

[illegible]

You are about to be asked to enter information that will be incorporated into your certificate request.

What you are about to enter is what is called a Distinguished Name or a DN.

There are quite a few fields but you can leave some blank

For some fields there will be a default value.

If you enter '.', the field will be left blank.

```
Country Name (2 letter code) [AU]:MX
State or Province Name (full name) [Some-State]:Coahuila
Locality Name (eg, city) []:Saltillo
Organization Name (eg, company) [Internet Widgits Pty Ltd]:TecNM
Organizational Unit Name (eg, section) []:localhost
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:
```

```
roberto@Debian-Server:~$
```

Se configuró HTTPS mediante un certificado autofirmado, logrando cifrar la comunicación entre cliente y servidor.

## Paso 6. Firewall

```
roberto@Debian-Server:~$ sudo ss -tuln
Netid  State  Recv-Q  Send-Q               Peer Address:Port              Local Address:Port
udp    UNCONN  0        0               0.0.0.0:*              10.0.2.15:68
udp    UNCONN  0        0               0.0.0.0:*              0.0.0.0:45135
udp    UNCONN  0        0               0.0.0.0:*              0.0.0.0:5353
udp    UNCONN  0        0               0.0.0.0:*              [::]:5353
udp    UNCONN  0        0               [::]:*
[fd17:625c:f037:2:9960:8e62:bc:9c3]:546
[::]:*
[fe80::8f7e:3346:c452:c0ca]%enp0s3:546
[::]:*
[::]:36410
tcp    LISTEN  0        80               0.0.0.0:*              127.0.0.1:3306
tcp    LISTEN  0       4096               0.0.0.0:*              127.0.0.1:631
tcp    LISTEN  0        511               0.0.0.0:*              *:80
tcp    LISTEN  0       4096               *:80
[::]:631
[::]:*
tcp    LISTEN  0        511               *:80
*:443
*:80
*:443

roberto@Debian-Server:~$ 
roberto@Debian-Server:~$ sudo ufw status
Status: active

To Action From
--
80 ALLOW Anywhere
443 ALLOW Anywhere
80 (v6) ALLOW Anywhere (v6)
443 (v6) ALLOW Anywhere (v6)

roberto@Debian-Server:~$
```

Se configuró el firewall permitiendo únicamente los puertos necesarios (80 y 443).

## Paso 7. Logs

```
roberto@Debian-Server:~$ sudo tail -f /var/log/apache2/access.log
127.0.0.1 - - [20/Mar/2026:07:13:48 -0600] "GET /repositorio/js/jquery.dataTables.js HTTP/1.1" 200 113714 "http://127.0.0.1/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [20/Mar/2026:07:13:50 -0600] "GET /repositorio/fonts/glyphicons-halflings-regular.woff2 HTTP/1.1" 200 18316 "http://127.0.0.1/repositorio/css/bootstrap-3.3.5.min.css" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
::1 - - [20/Mar/2026:07:13:53 -0600] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.66 (Debian) (internal dummy connection)"
::1 - - [20/Mar/2026:07:13:56 -0600] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.66 (Debian) (internal dummy connection)"
127.0.0.1 - - [20/Mar/2026:07:21:06 -0600] "GET / HTTP/1.1" 200 478 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [20/Mar/2026:07:21:06 -0600] "GET /favicon.ico HTTP/1.1" 404 527 "http://localhost/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [20/Mar/2026:07:27:35 -0600] "GET / HTTP/1.1" 200 462 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [20/Mar/2026:07:52:00 -0600] "GET / HTTP/1.1" 200 462 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [20/Mar/2026:07:53:05 -0600] "GET / HTTP/1.1" 200 3240 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [20/Mar/2026:07:53:05 -0600] "GET /favicon.ico HTTP/1.1" 404 458 "https://localhost/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"

roberto@Debian-Server:~$ sudo tail -f /var/log/apache2/error.log
[Fri Mar 20 07:58:46.971864 2026] [ssl:warn] [pid 3947:tid 3947] AH01906: Debian-Server.myguest.virtualbox.org:443:0 server certificate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Mar 20 07:58:46.971915 2026] [ssl:warn] [pid 3947:tid 3947] AH01909: Debian-Server.myguest.virtualbox.org:443:0 server certificate does NOT include an ID which matches the server name
[Fri Mar 20 07:58:46.972184 2026] [mpm_prefork:notice] [pid 3947:tid 3947] AH00163: Apache/2.4.66 (Debian) OpenSSL/3.5.4 configured -- resuming normal operations
[Fri Mar 20 07:58:46.972189 2026] [core:notice] [pid 3947:tid 3947] AH00094: Command line: '/usr/sbin/apache2'
```

Se verificó el monitoreo del servidor mediante los logs de acceso y errores.

## PASO 8 – Checklist

- ✓ Apache funcionando
- ✓ Página visible
- ✓ Seguridad aplicada
- ✓ Módulos deshabilitados
- ✓ HTTPS activo
- ✓ Firewall configurado
- ✓ Logs funcionando

## CONCLUSIÓN

En esta práctica pude comprender cómo mejorar la seguridad de un servidor Apache mediante diferentes configuraciones. Aprendí que no solo basta con instalar un servicio, sino que es importante protegerlo ocultando información sensible, deshabilitando funciones innecesarias y utilizando HTTPS para cifrar la información. También entendí la importancia del uso de firewall y monitoreo mediante logs. En general, esta práctica me ayudó a entender mejor cómo aplicar medidas de seguridad básicas en servidores web y su importancia en entornos reales.